

Design and Fabrication of Multirotor UAV for Medical Applications

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Abstract— Majority of India's 1.4 billion citizens are served by roughly 30,000 government-run primary healthcare centres, but at least 5-10% of them are inaccessible to medical suppliers owing to geographic locations and natural disasters. The COVID-19 pandemic has revealed numerous gaps in health-care systems around the world. Global shortages of essential supplies have resulted from supply chain issues. A Community's ability to prevent, detect, and respond to the disease has also been defined by timely access to testing and vaccination. Health care inequities frequently force members of rural and remote indigenous communities to travel to off-reserve locations, which delays access, is burdensome and costly. This project presents the design and fabrication of Unmanned Aerial Vehicles (UAV) for medical application. The design process starts with defining the performance and mission requirements including the types of configurations, frame type, payload and mechanism used. Nowadays UAVs are common, the mechanism used to drop the payload for medical application is complicated for different regions, we mainly focused to design the payload mechanism for medical application. After completing the design, the mechanism will be manufactured using composite materials. Flight tests will be conducted to evaluate the UAV's performance, dynamic characteristics and to demonstrate the success of the design. Here, multirotor is employed for the medical application. An autonomous mission can be planned with mission planner software, and changes can be brought on the flight if needed. This will help with detailed indoor inspections in a relatively less period.

Keywords—UAV, Multirotor, Medical application, Hexacopter, Fabrication, Payload, Flight time.

I. INTRODUCTION

A drone generally refers to any unmanned vehicle but technically it is an unmanned aircraft. Drones are more formally called unmanned aerial vehicles (UAV). An Unmanned aircraft system (UAS) is the entire system that comprises of UAV, Ground control system, software, etc. To be precise, a drone is a flying robot that can be remotely controlled or fly autonomously through software-controlled flight plans in their embedded systems [1,2]. It works in conjunction with onboard sensors and GPS. Initially drones were used in the military for information gathering and as a weapon. But now drones are being used in various fields like logistics, weather monitoring, photography, agriculture, and so on. The next human era begins with the advancement of UAVs in various applications [3-5].

One of its major applications is in logistics. A Multirotor UAV suits best for its application due to its of carrying heavy payloads for higher range. As these UAVs does not need any big runways for operations, it cuts the need of big infrastructure. It can perform autonomous missions without intervention or need of any support. It can reach the destination much faster as it moves completely air medium, less traffic, and less obstacles. Drone delivery will impact the

transportation of goods. The use of a drone delivery system could shrink delivery times from a matter of days to a matter of minutes [6].

Multirotor drones are the most common types of drones, which use more than two rotors to generate lift. The Multirotor drones have three, four, six, or eight rotor blades. The number of rotors in a drone is directly proportional to the thrust produced by it. We can change the speed of the rotors as greater than, less than, or equal to the forces acting on the drone [7,8]. This will make the drone ascend, descend, or hover accordingly. Also, if we change the speed of individual rotors, we will be able to roll, yaw the drone. One of the main reasons for the success of multirotor drones is that it is easy to practice and master it. It gives great control over the position and is the easiest option for getting an eye in the sky. Heavy lift drones such as drones used in industrial sectors for lifting heavy payloads, heavy-duty cameras, delivery drones will mostly be octocopters or Hexacopter.

The downside of the multirotor drone is they tend to spend most of the battery source on stabilizing themselves in the air. Often with a minimal payload like cameras, the flight time of most of the multirotor drones is 20-30 minutes. The throttle is changed to keep them stabilized in high precision. This is the reason behind us sticking to an electric motor and not using a gas engine to power the multirotor. Despite being relatively cheap and easy to handle, they have limited endurance. The flying time and speed are also less. So, they are not suitable for long-distance aerial mapping, long-duration surveillance, monitoring, etc. [9].

If we look at the idea of medicine delivery services in terms of drone technology, we will think about drones with specialized boxes which can carry the medicines to hospitals and rescue areas. If the person in rural or urban area needs medical equipment's within a very short period of time, due to traffic and other obstacles the things can't be delivered on time using road transport vehicles. The drone must be in a hover condition to drop the medicines or surgical products in hospital or rescue areas. The solution should be drop the payload in an exact position at the hovering condition of the drone, with a calculated flight time.

Steps involved in delivery process:

- In case of any medical emergencies the user can avail this service by placing a request to concern hospitals and laboratories, the UAV will reach the concern facilities to pick up the medical utilities.
- Once the facility approves for the delivery, the UAV will takeoff from the facilities to the desired user's location using GPS Coordinates. The UAV and autonomous flight software will take care of the mission.